

Bugs found: 4

1. The return value of malloc for pointer 'p' is not checked for NULL before dereferencing.
2. The pointer 'q' is used to access memory (q[2]) after the base pointer 'p' has been freed (Use-After-Free).
3. The access q[2] is an out-of-bounds memory access because only space for two integers was allocated.
4. The memory allocated to pointer 'x' is never freed, resulting in a memory leak.